Aishwarya Ravindra Fursule

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Summary

I am a Ph.D. student who loves building, testing, and questioning AI systems, especially in audio and speech. My work explores deepfake detection, speech foundation models, Audio LLMs, multimodal learning, fairness, and robustness. I enjoy connecting ideas across fields, trying things that may or may not work, and collaborating with people who think differently. I am always open to research, industry, internship, and collaboration opportunities—because good ideas become much more interesting when we build them together.

Education

Ph.D. in Computer Science, Wichita State University, Wichita, KS, USA *08/2024 – 12/2027 (Expected)*
M.Tech. in Mechanical Engineering, Cummins College of Engineering for Women, Pune, India *08/2017 – 07/2019*
B.E. in Mechanical Engineering, North Maharashtra University, Maharashtra, India *07/2013 – 05/2017*

Technical Skills

Programming & Systems: Python, MATLAB, C, Bash, Git, Linux, Conda, Slurm, CUDA, HPC, LaTeX

Machine Learning & Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn, CNN/ResNet, BiLSTM, Transformers, Transfer Learning, Multimodal Learning, Model Calibration, Statistical Evaluation

Speech & Audio AI: WavLM, HuBERT, wav2vec 2.0, XLS-R, Audio LLMs, ASR, MFCC, LFCC, Mel/Log-Spectrograms, openSMILE, Temporal Modeling

Research Methods: Audio Deepfake Detection, Temporal Localization, Fairness Evaluation, Distribution-Shift Analysis, Cross-Dataset Validation, Ablation Studies, Statistical Testing, Reproducible ML Pipelines

Research Experience

Wichita State University Wichita, KS, USA
Graduate Research Assistant – Machine Learning, Audio AI & Multimodal Learning *08/2024 – Present*

- Develop end-to-end PyTorch pipelines for audio deepfake detection and temporal localization, including preprocessing, frame-level labeling, feature extraction, model training, calibration, and cross-dataset evaluation on Linux/Slurm GPU clusters with CUDA.
- Fine-tune and benchmark speech foundation models including WavLM, HuBERT, wav2vec 2.0, and XLS-R against CNN/ResNet and sequence-model baselines across datasets, generators, languages, and distribution shifts.
- Investigate Audio LLMs and multimodal/multi-view learning by combining raw and spectral audio, self-supervised speech embeddings, ASR-derived text, and audio-language representations for manipulated-speech detection and localization.
- Study fairness, bias, robustness, and model calibration in audio AI by analyzing subgroup disparities, training-set composition, attack/edit types, recording conditions, and out-of-distribution performance.
- Conduct phoneme- and edit-level analysis of real and synthetic speech to identify acoustic and linguistic patterns associated with deepfake artifacts and model prediction behavior.
- Build and evaluate multilingual and phonetically annotated speech benchmarks and perform statistical, generator-wise, cross-dataset, and out-of-distribution analyses with research collaborators across institutions.

Wichita State University Wichita, KS, USA
Graduate Teaching Assistant – Machine Learning *08/2024 – 05/2025*

- Supported machine learning labs and student projects through implementation guidance, debugging, experimentation, and evaluation.

Industry Experience

Tata Motors Limited, Engineering Research Centre Pune, India
Senior Manager, Engine Design & Development *08/2019 – 12/2023*

- Owned engine component design and development across the full product lifecycle, coordinating design, testing, manufacturing, and supplier teams to meet program release milestones.
- Analyzed engine test, vibration, validation, and field-performance data to support design decisions and resolve durability and product-development issues.
- Led technical design reviews and produced engineering documentation for cross-functional teams and suppliers in a regulated automotive development environment.

Publications

1. **A. Fursule**, V. Nallaguntla, S. Kshirsagar, and A. R. Avila, "What You Train Is What You Get: Gender Bias, Training Composition, and Post-Hoc Mitigation in Audio Deepfake Detection," ***IEEE Transactions on Reliability***, 2026. *(Under review)*

2. Y. Tseng, **A. Fursule**, A. Z. Ma, V. Nallaguntla, A. Avila, S. Kshirsagar, and D. Harwath, "SPADE: A Multilingual Dataset for Speech Partial Deepfake Detection and Localization," ***IEEE Spoken Language Technology Workshop (SLT)***, 2026.

3. **A. Fursule**, S. Kshirsagar, and A. R. Avila, "Gender Fairness in Audio Deepfake Detection: Performance and Disparity Analysis," ***IEEE Conference on Artificial Intelligence (CAI)***, 2026.

4. **A. Fursule**, S. Kshirsagar, and A. R. Avila, "Towards Trustworthy Audio Deepfake Detection: A Systematic Framework for Diagnosing and Mitigating Gender Bias," ***IEEE International Conference on Systems, Man, and Cybernetics (SMC)***, 2026.

5. V. Nallaguntla, **A. Fursule**, S. Kshirsagar, and A. R. Avila, Naturalness Evaluation," ***Language Resources and Evaluation Conference (LREC)***, 2026.

6. D. E. Temer, A. Hamadene, M. S. Allili, V. Nallaguntla, **A. Fursule**, S. Kshirsagar, and A. R. Avila, "Phonetic Analysis of Real and Synthetic Speech Using HuBERT Embeddings: Perspectives for Deepfake Detection," ***IEEE International Conference on Systems, Man, and Cybernetics (SMC)***, 2025.

Leadership, Outreach & Service

- Member, IEEE Systems, Man, and Cybernetics Society, 2026
- Graduate Student Member, IEEE
- Workshop Instructor, IFML Workshop for High School Students, Wichita State University, June 2026
- Volunteer, IFML Workshop for High School Students, Wichita State University, June 2025
- Volunteer, IEEE GreenTech Conference, Wichita State University, 2025
- Volunteer, Club Parkinson's, Wichita State University, 2026